SYNAPSE OS: MACHINE LEARNING ESSENTIALS

Lecture Notes & Research Summary

1. NEURAL NETWORKS & BACKPROPAGATION

A neural network is a collection of connected units or nodes called artificial neurons.
Each connection transmits a signal, which is a real number computed by an activation function.

2. LOSS FUNCTIONS & GRADIENT DESCENT

Mean Squared Error (MSE): L = 1/N * sum((y_true - y_pred)^2)
Cross-Entropy Loss: L = -sum(y_true * log(y_pred))
Optimization: theta = theta - lr * grad(L)

3. REGULARIZATION TECHNIQUES

- L1 Regularization (Lasso): Promotes sparsity in weights
- L2 Regularization (Ridge / Weight Decay): Prevents extreme weights
- Dropout: Randomly zeros activations during forward pass with probability p

4. TRANSFORMER ARCHITECTURE

Self-Attention mechanism: Attention(Q, K, V) = softmax(Q * K^T / sqrt(d_k)) * V
Multi-Head Attention allows the model to jointly attend to information from different representation